GreenTrace

Indian College Student Carbon Footprint Tracker

Final Project Documentation

Project Name	GreenTrace — Indian Product Carbon Footprint Tracker
Type	1M1B Internship Project — Final Submission
Team	Prakruthi BR | Keerthi N | Pragathi BR | Bhumika R
Institution	Alliance University
Year	2026
Prototype	https://splendid-fudge-521fc5.netlify.app/

Tracing the Carbon Footprint of Everyday Indian Products

1. Executive Summary

GreenTrace is an interactive Environmental Life Cycle Dashboard built specifically for Indian college students. It traces the hidden environmental cost of 20 everyday products used in Indian college life — from smartphones and Maggi noodles to thermocol plates and shampoo sachets — across five life cycle stages, and guides students toward local Indian eco-friendly alternatives.

Unlike general environmental dashboards built by other teams, GreenTrace is grounded entirely in the Indian context. It uses local data from CPCB (Central Pollution Control Board), Indian manufacturer sustainability reports and Indian waste statistics. It also includes a unique decision-making layer called Should I Switch? that gives every user a personalised recommendation based on the product impact score.

2. Problem Statement

2.1 The Challenge

Indian college students use 30 to 50 single-use plastic items every single day. Most are unaware of the environmental cost of the products they buy. No India-specific interactive tool existed to compare the sustainability of products side by side in a way that is relevant to college life.

2.2 Why GreenTrace is Different

Our faculty mentor pointed out that multiple teams explored similar environmental dashboard ideas. GreenTrace addresses this by differentiating on three key dimensions:

- Indian context — all 20 products are things Indian college students actually use daily, with India-specific data
- Local alternatives — eco-friendly options are Indian brands like Khadi, Patanjali, Bamboo India, and traditional practices like Kabadiwala recycling
- Decision-making layer — a Should I Switch? recommendation tells students whether switching is high priority, worth considering, low priority or already a good choice

2.3 Environmental Impacts Addressed

- CO_2 Emissions — direct driver of climate change and air pollution in Indian cities
- Water Usage — critical in India where 163 million people lack access to clean water
- Energy Consumption — India is still heavily dependent on coal-based electricity
- Waste Generation — India generates 9.46 million tonnes of plastic waste per year

3. Project Objectives

1. Analyze 5 life cycle stages for 20 Indian college student products
2. Measure CO_2 emissions, water usage, energy consumption and waste generation per stage

3. Ground all data in Indian context using CPCB data, Indian brand reports and Indian waste statistics
4. Compare products side by side to help students identify their highest-impact daily items
5. Add a Should I Switch? decision-making layer to guide each student toward specific actions
6. Provide local Indian eco-friendly alternatives for every product analyzed
7. Build an interactive dashboard that works on both laptop and mobile devices

4. What Makes GreenTrace Unique

Feature	Other Teams	GreenTrace
Focus	General products globally	Indian college student products specifically
Data	Generic global LCA data	CPCB, Indian manufacturer and Indian waste data
Alternatives	Generic eco options	Local Indian brands — Khadi, Patanjali, Bamboo India, Kabadiwala
Decision Layer	None	Should I Switch? — High Priority, Consider, Low Priority, Good Choice
Indian Context	No local relevance	Bengaluru drainage, India plastic ban 2022, hostel life, sachets
Products	Generic 10 items	20 items students actually use — Maggi, sachets, thermocol plates

5. Methodology

5.1 Life Cycle Stages Analyzed

Stage	Description	Typical Share of Impact
Raw Materials	Mining, agriculture, petroleum refining and forestry	30 to 45 percent
Manufacturing	Assembly, chemical processing and packaging	25 to 50 percent
Distribution	Transport from factory to consumer via road and rail	5 to 15 percent
Use	Environmental impact during the product active life	5 to 20 percent
Disposal	Landfill, recycling, incineration or composting	3 to 10 percent

5.2 Metrics Tracked

Metric	Unit	What It Measures	Indian Relevance
CO_2 Emissions	kg CO_2e	Total greenhouse gases across full life cycle	India is 3rd largest CO_2 emitter globally
Water Usage	Litres	Total fresh water consumed	163 million Indians lack clean water
Energy Use	kWh	Total energy from extraction to disposal	India is 70 percent coal dependent
Waste Generated	kg	Solid waste across full life cycle	India generates 9.46 MT plastic waste yearly

5.3 Impact Scoring System

Impact Level	Score Range	Description and Recommendation
High Impact	70 to 100	Severe environmental damage. Should I Switch says HIGH PRIORITY — switch immediately
Medium Impact	50 to 69	Moderate impact. Should I Switch says CONSIDER SWITCHING
Low Impact	35 to 49	Below average impact. Should I Switch says LOW PRIORITY
Very Low	0 to 34	Minimal impact. Should I Switch says GOOD CHOICE — keep using it

5.4 Decision-Making Layer — Should I Switch?

Every product in GreenTrace includes a personalised Should I Switch? recommendation displayed at the bottom of the product detail panel. This is the feature that differentiates GreenTrace from all other environmental dashboards. The recommendation uses the impact score to give students one clear action:

- Score 70 and above — HIGH PRIORITY shown in red. Switch this product as soon as possible. It will make a significant difference to your carbon footprint.
- Score 50 to 69 — CONSIDER SWITCHING shown in orange. This product has moderate impact. Small changes here add up over time.
- Score 35 to 49 — LOW PRIORITY shown in yellow. This product has below-average impact. Focus on higher-impact items first.
- Score below 35 — GOOD CHOICE shown in green. This product has very low environmental impact. Keep using it.

6. Products Analyzed (20 Indian College Student Products)

#	Product	Category	Score	Level	Key Indian Issue
1	Smartphone	Electronics	95	High	22% e-waste recycled in India
2	Plastic Water Bottle	Packaging	85	High	Clogs Bengaluru drainage
3	Thermocol Plate	Packaging	88	High	Banned in India 2022, still at fests
4	Earphones / Earbuds	Electronics	82	High	Fast fashion electronics, 1 year life
5	Maggi Instant Noodles	Food	70	High	Multi-layer foil, top hostel meal
6	Plastic Tea Coffee Cup	Packaging	72	High	Every college canteen, non-recyclable
7	AA and AAA Batteries	Electronics	65	Medium	Toxic metals in Indian landfills
8	Plastic Carry Bag	Packaging	68	Medium	Banned but still widely used
9	Plastic Food Container	Packaging	58	Medium	BPA leaching in hostel microwaves
10	Shampoo Sachet	Personal Care	60	Medium	8 billion sachets per year in India
11	Plastic-Covered Notebook	Stationery	62	Medium	Plastic cover is non-recyclable
12	Face Cream Tube	Personal Care	55	Medium	Laminate tube rejected by recycling
13	Disposable Razor	Personal Care	50	Medium	2 billion discarded in India yearly
14	Plastic Toothbrush	Personal Care	48	Low	3.5 billion discarded in India yearly
15	Tissue Paper Pack	Household	45	Low	Replacing traditional Indian cloth
16	Liquid Soap Bottle	Personal Care	42	Low	HDPE recyclable via kabadiwala
17	Plastic Soap Bottle	Personal Care	38	Low	HDPE recyclable but rarely is
18	Amazon Flipkart Box	Packaging	35	Low	India delivers 10M+ packages daily

#	Product	Category	Score	Level	Key Indian Issue
19	LED Light Bulb	Household	28	Very Low	Good choice, improper disposal issue
20	Ball Point Pen	Stationery	25	Very Low	440 million pens discarded yearly

7. Stakeholders

Stakeholder	Role and Contribution
Indian College Students	Provided product usage data, validated Indian product selection, completed feedback survey
Project Team	Built dashboard, collected and analyzed life cycle data, created Should I Switch recommendations
Faculty Mentor	Guided research direction, reviewed methodology, gave critical feedback to differentiate the project
Community Members (Bengaluru)	Contributed survey responses on environmental awareness and eco product availability

7.1 Survey Results — Google Forms Round 1

Result	Finding
68%	Use plastic packaging products on a daily basis
75%	Prioritise price and convenience over sustainability when buying
52%	Were unaware of the full environmental impact of products they use daily
81%	Expressed interest in learning about eco-friendly Indian alternatives

8. Data Sources

- Google Forms Survey (Primary) — Survey responses from Indian college students on product usage patterns and environmental awareness. Used to validate all 20 product selections.
- CPCB — Central Pollution Control Board, India — Official Indian statistics on plastic waste generation, e-waste volumes and landfill data across Indian states.
- Online Environmental Research Reports — Peer-reviewed Life Cycle Assessment studies and sustainability reports from international and Indian environmental organisations.
- Kaggle — Global Plastic Waste 2023 Country-Wise Analysis — Used to contextualise India-specific packaging impact data and validate waste generation figures.
- Indian Brand Sustainability Reports — HUL, Patanjali, ITC, Nestlé India, Colgate India annual sustainability disclosures and product environmental data.

- CSE India (Centre for Science and Environment) — India-specific reports on plastic pollution, urban waste and consumer product environmental impact.

9. Dashboard Features

9.1 Pages

- Dashboard — KPI cards, life cycle stage flow, key metric overview and live data indicator
- Products — Search bar, 6 category filters, 20 product cards each showing impact badge, score, CO_2, water, energy and waste data
- Compare — Select up to 4 products for side-by-side comparison table with Best badges on lowest-impact values
- Insights — Interactive charts with metric filter, product filter and Bar, Line, Pie and Donut chart type switcher
- About — Project background, survey results with stats, and full team information

9.2 Key Features

- Impact Level Badges — High, Medium, Low and Very Low shown on every product card and in the comparison table
- Should I Switch? — Decision-making layer giving personalised recommendation on every product
- Life Cycle Breakdown — Per-stage CO_2, water, energy and waste data across all 5 stages for every product
- Indian Eco-Alternatives — 3 local Indian alternatives per product including Khadi, Patanjali, Bamboo India, Kabadiwala options
- Recyclability Index Chart — Percentage recyclable at end of life for all 20 products
- CO_2 Savings Donut Chart — Estimated percentage CO_2 reduction from switching to eco alternatives
- Percentage Labels — Shown outside each pie and donut chart slice for easy reading
- Horizontal Navbar — Clean top navigation with Dashboard, Products, Compare, Insights and About links
- Dark Green Footer — Team members listed with project details

10. Project Roadmap

Week	Phase	Activity	Status
1	Problem Definition	Identified Indian college student products and defined problem scope	Completed
2	Data Collection	Collected Indian-specific LCA data, CPCB statistics and ran Google Forms survey	Completed
3	Analysis	Processed data, calculated impact scores, built initial dashboard	Completed

Week	Phase	Activity	Status
4	Build and Test	Added decision-making layer, Indian alternatives, tested with 42 stakeholders, collected feedback	Completed

Estimated Project Duration: 30 to 40 Hours

11. Impact Measured

Metric	Result
Total stakeholders tested	42 students, faculty and community members
Awareness increase	76 percent reported higher LCA knowledge after using GreenTrace
Behaviour change	5 users confirmed they switched to at least one eco-alternative
Average satisfaction rating	4.2 out of 5 across all stakeholder groups
Would recommend GreenTrace	89 percent said definitely yes
Estimated CO_2 saved (100 users switching to reusable bottle)	Approximately 1,800 kg CO_2 per year
Plastic bags eliminated (100 users switching)	Approximately 5,200 bags per year

12. Project Team

Team Member	Role
Prakruthi BR	Research and Data Collection — gathered Indian-specific LCA and CPCB data
Keerthi N	Dashboard Development — built the interactive GreenTrace prototype
Pragathi BR	Life Cycle Analysis — analyzed all 20 products across 5 stages
Bhumika R	Presentation and Insights — created charts, insights and final report

GreenTrace

Tracing India's Product Carbon Footprint — One Student at a Time

1M1B Internship Project 2026 | Prakruthi BR | Keerthi N | Pragathi BR | Bhumika R